# USB POWER 45W S8.3 / SINGLE BOARD, +5V SERIES PROTECTION

S8.3: +5V fast series disconnect; +/−12V directly from converters; ceramic banks; two parallel outputs; VPD−powered OV guard; MAIN LED only. No crowbar.

## USB PD, main eFuse and VIN damper

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A common input switch
VIN damper 1R + 2 x 10u

File: 01_usb_input.kicad_sch

## +12 V converter

MAIN_OK
VIN
+12V

TPS62933F
+12 V delivered rail

File: 02_positive_12v.kicad_sch

## +5 V converter

MAIN_OK
VIN
P5_RAW

TPS62933F
P5_RAW before disconnect

File: 03_positive_5v.kicad_sch

## −12 V inverting converter

+12V
VIN
−12V

LMR38020F inverting buck−boost
−12 V delivered rail

File: 04_negative_converter.kicad_sch

## Two output headers − monitors, overvoltage guard and MAIN LED

+12V
−12V
MAIN_LED_OK
P5_RAW
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK
OV_N

Two Eurorack outputs J2 + J3
Monitors, OV guard, +5V eFuse
MAIN LED (only status LED)

File: 06_outputs_status.kicad_sch

FAULT POLICY
Persistent bad rail/status after ~0.57 s −> all OFF until RESET.
Severe OV via U20 acts even before ARMED.
U23 independently cuts +5V; OVLO alone is auto−recovering.
MAIN indicates enabled/status, not precision regulation.

OUTPUTS: +12 / +5 / −12 on J2 and J3 in parallel
+5V has U23 current limit (~3.03 A nominal), +/−12V retain converter−only limits.
Targets 2.5 / 2.5 / 1.0 A; not a qualified simultaneous rating.

## Always−on control − fault latch, overvoltage preset and RESET

MAIN_OK
N12_OK
OV_N
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always−on LDO / retained ARMED + FAULT
Manual RESET; OV_N presets FAULT

File: 05_fault_reset.kicad_sch

Crowbar removed; U18/U20 on VPD; split +5V banks; see docs/s8−single−board.md
S8.3: single board, +5V fast series eFuse; +/−12V unswitched; two headers; MAIN only

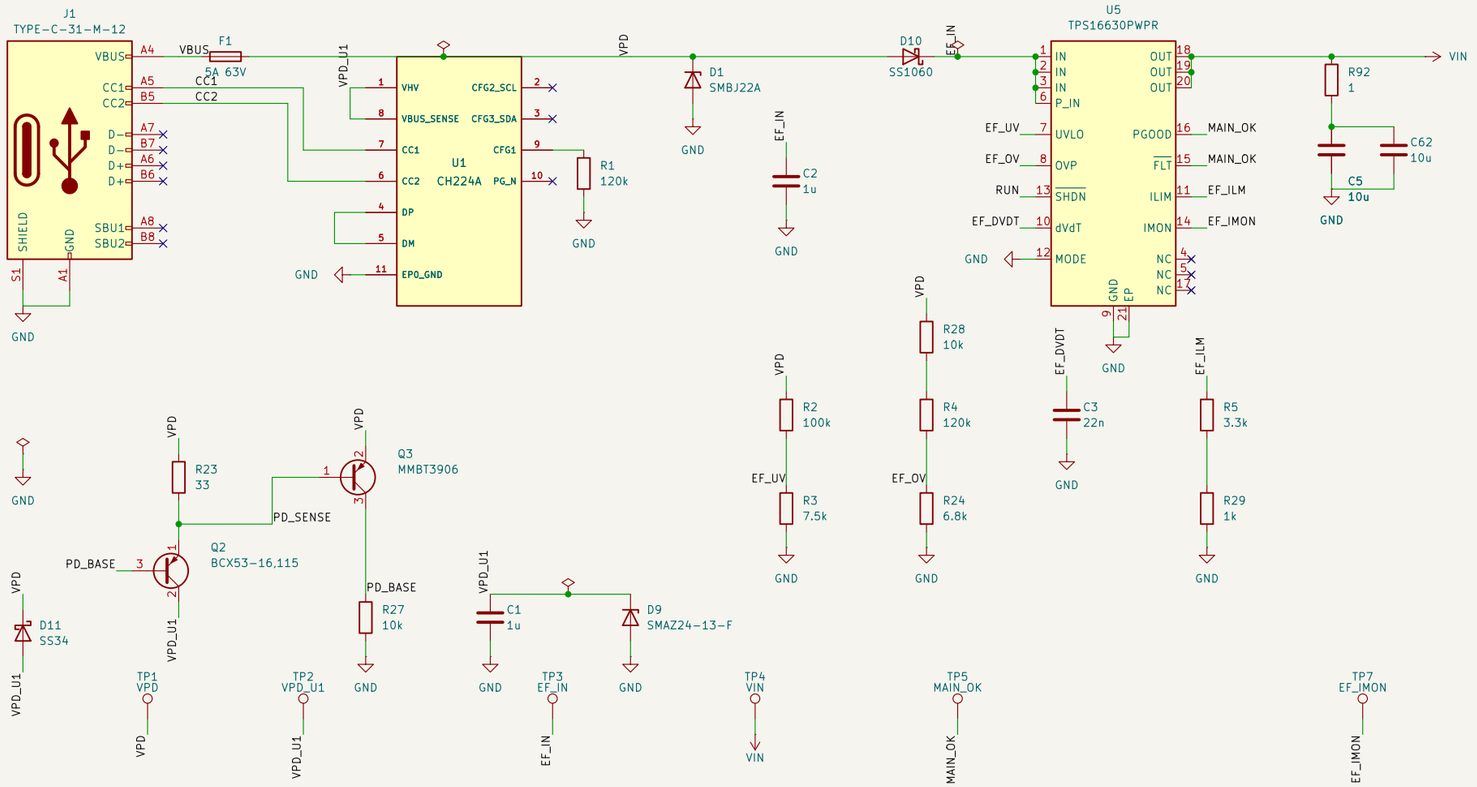
**Big Modular**

Sheet: /
File: usb−power−45w−single.kicad_sch

***Title: USB power S8 single board − wired power flow overview***

| Size: User | Date: | **Rev: S8.3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/7 |

# USB PD, MAIN EFUSE AND VIN DAMPER




20 V PD request: U5 TPS16630 (60 V, 4.186 A nominal) is the common input switch; RUN low turns every rail off.
MAIN_OK = FLT + PGOOD at 3.3 V. R92 1R + C5/C62 2 x 10u damp the all-ceramic VIN bus (replaces the 47u can).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK
RUN
VIN
VPD

S8 single board, S8.3 schematic (+5V series eFuse, VPD-powered comparators) – see docs/s8-singl

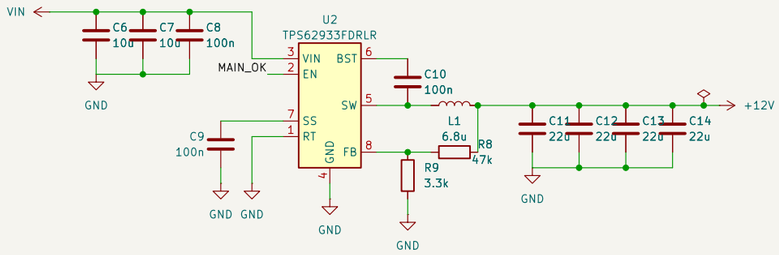
**Big Modular**

Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

***Title: USB PD, main eFuse and VIN damper***

| Size: User | Date: | **Rev: S8.3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/7 |

# +12 V CONVERTER







TPS62933F +12 V, unchanged power stage; its output IS the delivered +12V (no rail eFuse).
Soft start C9 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C11–C14 here + C34 (J2) and C59 (J3) on page 06; the 100u can is gone.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇—— +12V

MAIN_OK ◇—— MAIN_OK

VIN ◇—— VIN

S8 single board, S8.3 schematic (+5V series eFuse, VPD-powered comparators) – see docs/s8-singl

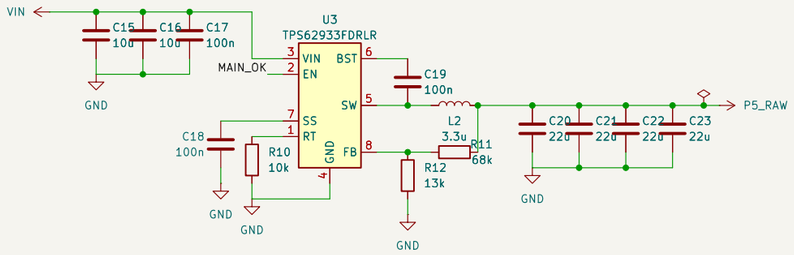
**Big Modular**

Sheet: /+12 V converter/
File: 02_positive_12v.kicad_sch

***Title: +12 V converter***

| Size: User | Date: | **Rev: S8.3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/7 |

# +5 V CONVERTER







TPS62933F +5 V: P5_RAW before U23. Feedback and C20–C23 remain upstream; never sense after U23.
C69–C72/C75/C76 extend the upstream bank to 10 x 22u. Delivered +5V has C35/C60/C73/C74 after U23.
C18 100n converter soft start; U23 has its own dVdt ramp. Converter/load stability is not bench qualified.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇—— MAIN_OK

P5_RAW ◇—— P5_RAW

VIN ◇—— VIN

S8 single board, S8.3 schematic (+5V series eFuse, VPD-powered comparators) – see docs/s8-singl

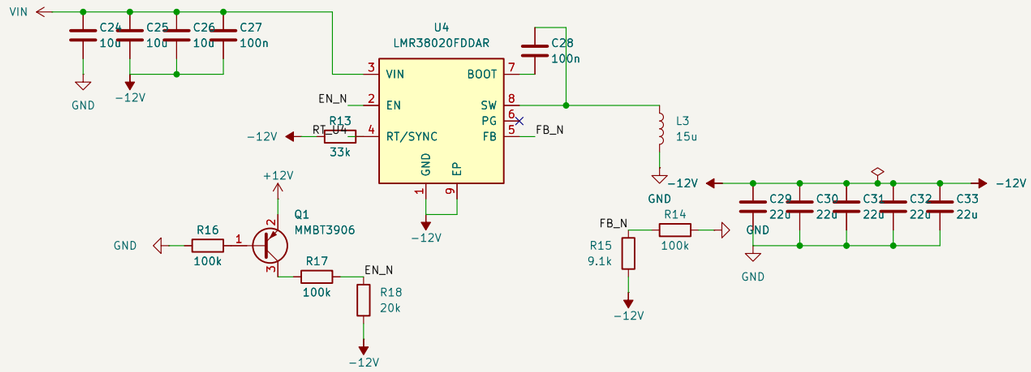
**Big Modular**

Sheet: /+5 V converter/
File: 03_positive_5v.kicad_sch

***Title: +5 V converter***

| Size: User | Date: | **Rev: S8.3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/7 |

# −12 V INVERTING CONVERTER







LMR38020F inverting buck−boost; IC ground / exposed pad = the delivered −12V (no series switch).
4 ms soft start, 18 ms OCP blanking at start−up. Q1 follows +12V. C29−C33 here + C36 (J2) and C61 (J3).

INTER−SHEET INTERFACE (see wired power−flow overview)

+12V ◇ +12V

−12V ◇ −12V

VIN ◇ VIN

S8 single board, S8.3 schematic (+5V series eFuse, VPD−powered comparators) − see docs/s8−singl

**Big Modular**

Sheet: /−12 V inverting converter/
File: 04_negative_converter.kicad_sch

***Title: −12 V inverting converter***

| Size: User | Date: | **Rev: S8.3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/7 |

ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET

U9
TPS7A1633DGNR
C46 1u
C47 10u
R63 10k
MAIN_OK

U10
TPS3808G01DBVR
R91 15k
R65 47k
R66 10k
R64 10k
R67 10k
R68 10k
C48 100n

U11
TPS3808G01DBVR
R69 33k
R70 10k
C49 100n
R71 10k
C50 100n

SW1
TL3340AF160QG

U12
SN74HC74DR

U13
SN74HC08DR

U14
SN74HC20DR
R72 10k
C51 100n

U15
SN74LVC1G17DBVR

C52 100n
C53 100n
C54 100n
C55 100n

TP9 RUN
TP10 ARMED
TP11 FAULT_LATCH
TP12 P12_OK
TP13 P5_OK
TP14 N12_OK
TP15 V20_OK
TP16 V3V3
TP6 OV_N

RESET clears ARMED and FAULT; RUN is forced low while reset is asserted.
Startup qualification ~0.57 s typical (C49 100n), then any persistent bad status latches every rail OFF until RESET.
OV_N (U20, page 06) presets FAULT at once, also inside the startup window. U14 gate 2 unused (inputs to GND).

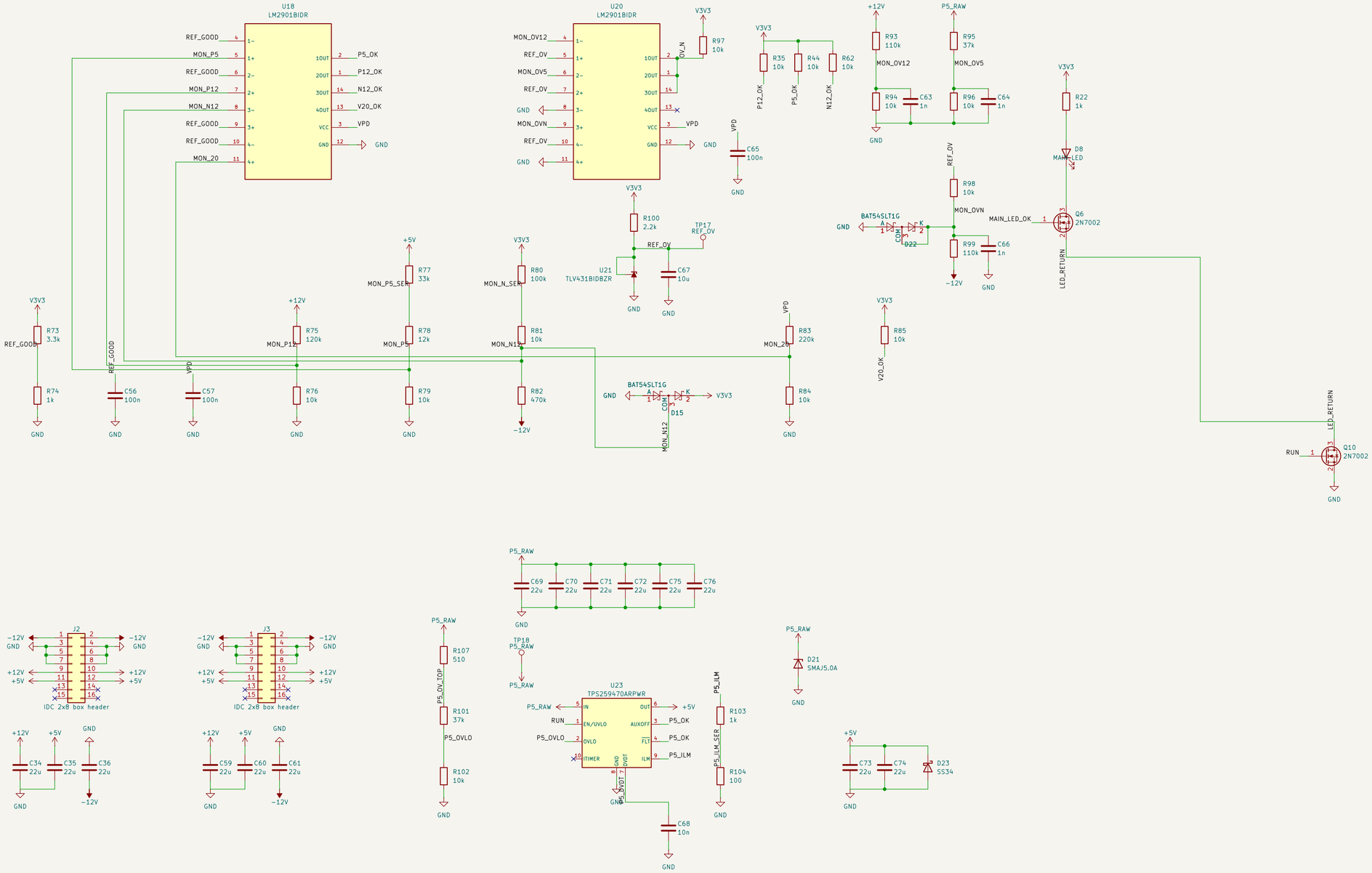
INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_LED_OK
P5_OK
MAIN_OK
RUN
N12_OK
V20_OK
OV_N
V3V3
P12_OK
VPD

S8 single board, S8.3 schematic (+5V series eFuse, VPD-powered comparators) — see docs/s8-singl
**Big Modular**
Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 05_fault_reset.kicad_sch
***Title: Always-on control — fault latch, overvoltage preset and RESET***
Size: User | Date: | **Rev: S8.3**
KiCad E.D.A. 10.0.6 | Id: 6/7

# TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

J2 + J3: two parallel 16-pin outputs, pin 1 = −12V. C34–C36 at J2, C59–C61 at J3 (22u each, replace the 100u cans).
D8 MAIN is the only status LED (rail LEDs removed): input-valid AND main-eFuse status AND RUN. Any bad rail after the
startup window or a U20 guard trip latches every rail off. Short OVLO-only pulses may recover. P12/P5/N12_OK: logic + test pads.
U20 overvoltage guard (S8.1): +12V > 14.88 V, +5V > 5.83 V or −12V < −13.64 V nominal (corners 14.48–15.27 / 5.68–5.98 /
13.27–14.02 V) pulls OV_N -> FAULT -> U5 off. Own reference REF_OV = U21 TLV431B (R100, C67, TP17); U18/U20 LM2901B:
R93–R96, R98, R99 are 0.1 % / 25 ppm. The −12V channel compares MON_OVN with GND (no V3V3 term); D22 clamps MON_OVN
at GND if V3V3 is lost while the rack holds −12V. D21 SMAJ5.0A is a transient clamp (6.4 V min breakdown), not an OV limit.
S8.3: U18/U20 supplies are always-on VPD; outputs remain 3.3 V logic, U20 +5V guard monitors P5_RAW.
U23 TPS259470A independently disconnects +5V at -5.70 V (1.2 us typical, no maximum specified). D21 moved to P5_RAW.
AUXOFF + FLT join P5_OK: faults latch after startup qualification. Severe raw OV is unmasked through U20.
Q11/Q12/Q13/U22 crowbar removed. No guaranteed 6 V peak or prototype qualification is claimed.







INTER-SHEET INTERFACE (see wired power-flow overview)

+12V — +12V
−12V — −12V
MAIN_LED_OK — MAIN_LED_OK
N12_OK — N12_OK
OV_N — OV_N
P12_OK — P12_OK
P5_OK — P5_OK
P5_RAW — P5_RAW
RUN — RUN
V20_OK — V20_OK
V3V3 — V3V3
VPD — VPD

S8 single board, S8.3 schematic (+5V series eFuse, VPD-powered comparators) — see docs/s8-singl
**Big Modular**
Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 06_outputs_status.kicad_sch

***Title: Two output headers — monitors, overvoltage guard and MAIN LED***

| Size: User | Date: | Rev: S8.3 |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 7/7 |